

LAB-1: PID control

1 Introduction

The main objective of this laboratory is to get a basic understanding for how feedback control can be used to modify the behavior of a dynamic system. In particular, we will consider PID-control of a simple process consisting of two water tanks. The letters *PID* stand for *Proportional*, *Integral*, and *Derivative* control, respectively. PID controllers are by far the most common type of controllers used in industrial systems, mainly because they are relatively simple and are often able to provide good performance. The laboratory experiments will hopefully confirm this.

In addition to establishing an understanding for the fundamental principles of feedback control, the laboratory will provide some experience of manual tuning of PID-controllers.

The process to be controlled consists of two water tanks and a pump (see picture below). Water is pumped from a reservoir to the upper tank, from which it flows through a restriction into the lower tank. From the lower tank the water flows through a second restriction back to the reservoir. The upper tank also has a tap from which, when opened, water is allowed to flow directly to the reservoir. The water levels in both tanks are measured using pressure sensors. The process and equipment are manufactured by Quanser Consulting in Canada. The control system is implemented in Matlab. The user interface is for the most part self-instructive.

This material is partly based on material from the Department of Automatic Control, Lund Institute of Technology, Sweden. This version of the manual was translated and revised by Jonas Wijk in 2001. A new revision was conducted in 2017 by Riccardo S. Risuleo and Miguel Galrinho, to comply with modifications in the implementation and user interface.

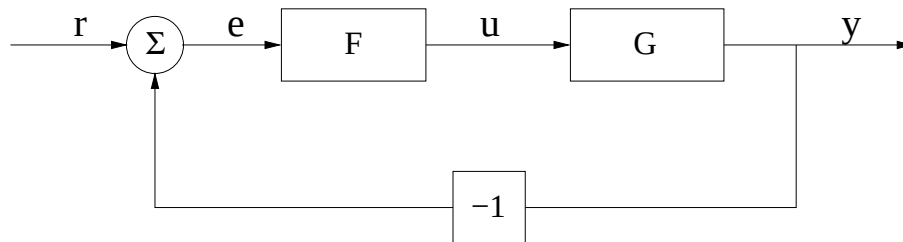


Coupled tanks with pump.

2 Preparations

2.1 Concepts and Notation

The block-diagram of a feedback control system is shown in the figure below.



Here, G is the system to be controlled, and F is the feedback controller. Referencing the block-diagram, briefly explain the following concepts:

Open-loop system:

.....

.....

Closed-loop system:

.....

.....

Output:

.....

.....

Reference value, or setpoint:

.....

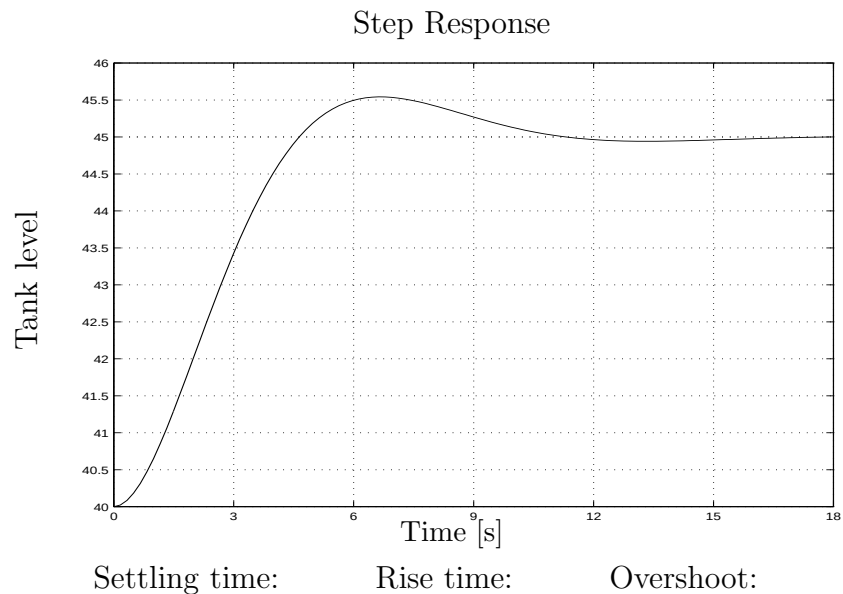
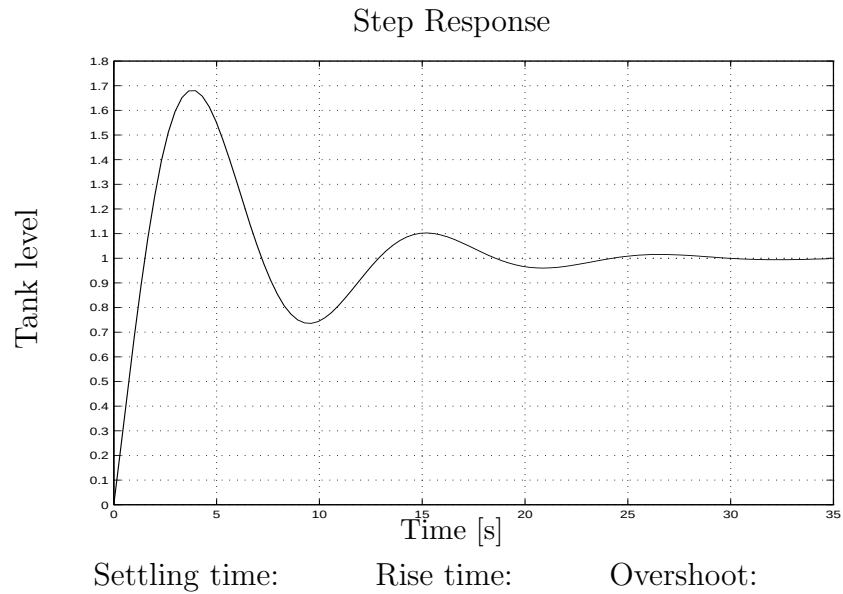
.....

Static error:

.....

2.2 Step Response Characteristics

In order to describe the performance of a control system, it is often useful to consider the step response of the system: that is, the response of the system output to a step in some input signal, such as the reference. Parameters that describe the step response are rise time, overshoot and settling time. Determine these quantities for the step responses below (hint for the bottom figure: the initial tank level is not zero!).



3 Manual control

In this exercise, you will control the water levels manually by observing the levels and adjusting the pump voltage accordingly. The aim is to investigate if the process is linear and to gain a feeling for how difficult it is to control the process.

How to start

1. Click twice on the Matlab icon on the desktop.
2. Wait until the Graphical User Interface is loaded.
3. In the **System settings** box, turn the second switch to **Real time**.
4. In the same box, turn the first switch to **Start**.

In the **Controller settings** box, choose **Manual** control.

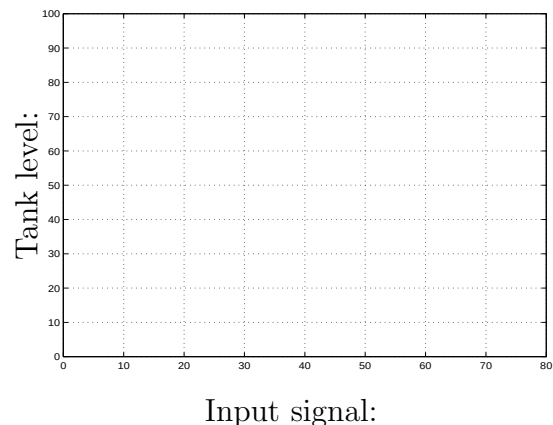
NOTE: All signals displayed are scaled from 0 to 100. An input signal of 0 gives 0V and 100 gives 15V. For the levels, a signal of 0 corresponds to 0cm and 100 to 25cm. *You should always use the scaled signals from 0 to 100!*

To more easily perform measurements, **Capture** the data to a MATLAB figure.

Task 3.1

- You will set the *level in the upper tank* to approximately 80 by *adjusting the input*. To adjust the input, you can use the sidebar **Input**. The upper tank is referred to as Tank1, so make sure that this signal is displayed by ticking the box **Tank1** under **Plot settings**. Let the system settle in a steady-state (i.e., constant levels). Write down the tank level and the corresponding input signal in the table below. Then, use the same procedure to change the level to approximately 40, let the system settle in steady-state, and write down the level and input signal. Plot the steady-state pairs (Input, Tank level) in the graph below.

Measurement:	Tank level:	Input signal:
1.	0	0
2.		
3.		



Is the process linear? Motivate!

.....

- Try to keep the level constant when somebody disturbs the system by opening the tap. Try to come back to the desired level as fast as possible when the tap is closed again.

Task 3.2

- Try now to adjust the level in the lower tank to 40 (make sure the **Tank2** level is displayed). Then, play around that value: for example, change it to 50, or try to maintain it when someone opens and closes the tap. Is this case more difficult? Does it take longer time? Motivate!

.....

.....

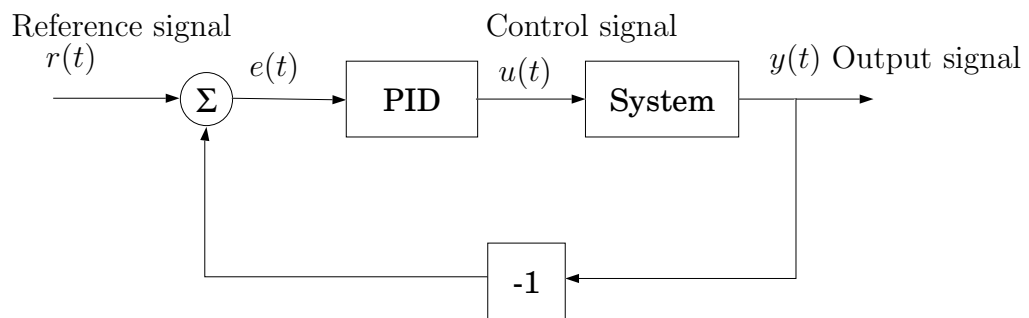
4 Proportional control

In the previous tasks, the process was controlled by manually adjusting the pump voltage. Through observations of the deviation from the desired level, you decided to increase or decrease the voltage. A controller makes such decisions automatically. In this laboratory, we will consider PID controllers, starting with the special case of pure proportional control, also known as P control.

PID controllers are determined by the three parameters K , T_I , and T_D . The values of these parameters will affect the behavior of the closed-loop system. At startup, the parameters should be $K = 2$, $T_I = \text{Inf } (\infty)$, and $T_D = 0$, which corresponds to a P-controller with $K = 2$. For pure P control, the relation between the input $u(t)$, reference $r(t)$ and output $y(t)$ is

$$u(t) = K[r(t) - y(t)] = Ke(t),$$

where $e(t)$ is called the *control error*.



Level control with PID controller

How to operate the PID controller and change the settings

1. Start the controller by turning the first switch under **Controller settings** to **PID**. Choose which tank that is to be controlled with the switch on the right.
2. To change a controller parameter, click on the box containing the parameter value and enter the new value.
3. Choose the reference signal for the tank level with the box **Reference** or the respective slider.

Task 4.1

- Choose PID-control of the upper tank (**Tank1**) with parameters $K = 2$, $T_I = \text{Inf}$ (∞), and $T_D = 0$. (make sure the **Tank1** signal is displayed).
- Set the reference value to $r=50$. Read the final level $y = \dots$ and the corresponding control signal $u \dots$. Calculate the static error $e=r-y=\dots$ and verify that the result agrees with the controller equation above, i.e., that $u = Ke$.
- Open the tank tap. Does the error change? Motivate!

.....

- Close the tap and increase the reference value to 70. Does the static error change compared to the case with $r = 50$? Motivate!

.....

- Close the tank tap and try to find a reasonable K -value fulfilling the following criteria:
 1. The static error should be as small as possible.
 2. The rise time and the settling time should be as short as possible.
 3. The control signal must not reach either limit value (0 or 100) for a change in the reference value from 40 to 45.

Contemplate what each of these criteria imply for the choice of K . Also note how the control signal becomes “noisy” for large values of K .

.....

An appropriate value of the controller gain is $K = \dots$

Why does the tank level never reach the reference value with the P-controller? (*Hint: what is the control signal if the control error is zero?*)

.....

Task 4.2

Now, you will control the lower tank, by turn the switch under **Controller settings** to **Tank2**. Here, you may need a longer time window: under the **Plot settings** box, you can change **T span** to, for example, 300 seconds.

Set the reference to 40 and wait for the signals to reach steady state with $K = 10$. Change the reference to 45 and measure rise time $T_r = \dots\dots\dots$, overshoot $M = \dots\dots\dots$, and settling time $T_s = \dots\dots\dots$

Try to find a reasonable value for K in the interval $2 \leq K \leq 10$, according to the three criteria in Task 4.1. Is it more difficult than for the upper tank? Motivate!

.....
.....

Task 4.3

Conclusions:

- Is P control satisfactory for the upper tank? Motivate!

.....

- Is P control satisfactory for the lower tank? Motivate!

.....

- What do you consider to be the main disadvantage with P-control?

.....

- What imposes the upper limit for the controller gain K when controlling the level in the upper tank? Use the criteria in Task 4.1.

.....

- What imposes the upper limit for K when controlling the level in the lower tank? Is it possible to satisfy all the criteria in Task 4.1?

.....

Task 4.4

Draw a block diagram for the closed-loop system for the case when the level in the lower tank is the controlled output. Represent each tank by a separate block, and indicate the variables *control input*, *upper tank level*, *lower tank level*, as well as the *disturbance* represented by opening the tap.

If any of the assistants are available, then this is a good opportunity for partial reporting of the investigation. Otherwise, continue working and report later.

Partial report accepted by:

5 PI control

With a PI controller, the relationship between the control error $e(t) = r(t) - y(t)$ and the control signal $u(t)$ is

$$u(t) = K[e(t) + \frac{1}{T_I} \int_0^t e(\tau) d\tau]$$

The parameters K and T_I can be changed on the computer screen. The integral part,

$$\frac{K}{T_I} \int_0^t e(\tau) d\tau \quad (1)$$

can be zeroed with the button **I Reset**.

Task 5.1

We first consider PI control of the level in the upper tank (make sure **Tank1** is selected under **Controller settings**). Tick also the boxes **P-part** and **I-part** in **Signals**, under **Plot settings**. This will separately display the “P part” $Ke(t)$ and the “I part” (1) of the control signal, respectively. Choose $K = 10$, $T_I = 20$ and set the reference value to 40. Verify that the tank level reaches the reference value (i.e., that the static error becomes zero). Then, open the tank tap and verify that the static error becomes zero again.

What are the sizes of the “P part” and the “I part” when the error has become zero? Compare with the input value found in Task 3.1. Motivate!

.....

.....

Task 5.2

Before continuing with more PI-control experiments, we will consider a particular problem that can arise when integral control is combined with control inputs that can saturate (i.e., reach a maximum or minimum limit). To understand it, we perform the following experiment (make sure the input signal, P-part, and I-part are displayed).

- Choose $K = 10$, $T_I = 2$, and select **Tank1** to be controlled. Turn the **Anti-windup** switch to **Off**. Set the reference value to 30, and wait until the level is constant. Then, change the reference value to 50. Determine the overshoot $M = \dots\dots$ and the settling time $T_s = \dots\dots$
- Repeat the experiment with the **Anti-windup** switch **On**. Determine the overshoot $M = \dots\dots$ and the settling time $T_s = \dots\dots$

In the first experiment, you can observe that the input (pump voltage) remained constant at its upper limit of 100 (15V) even after the level had reached its setpoint. This is caused by a phenomenon known as *integral windup*, which occurs when the real input is saturated. Although the real input is constrained, the computed input and its integral part continue to

grow, because the controller “believes” that the input should be larger. However, because of the saturation, this increase in the computed input has no effect. When the tank level becomes larger than the reference, the control error becomes negative, and the input must decrease to lower the tank level. While the proportional part of the input becomes negative instantaneously, the integral part may by then be very large. Then, an error of opposite sign has to be accumulated to bring back the integral part to its final value that drives the error to zero. This causes the overshoot to increase and a delay in the response.

A solution to the problem of integral windup is to prevent the integral part from growing when the control signal is saturated. This is what is implemented when **Anti-windup** is **On**. During the remaining part of the lab, the **Anti-windup** switch shall be *always On*.

Task 5.3

We now proceed to experiment with how PI control is appropriate for disturbance elimination, and how T_I affects the response time. We first do this for the upper tank (make sure that **Tank1** is controlled and displayed). Choose $K = 8$, $T_I = 20$ and set the reference value to 40 with the tap closed. Wait until the level is steady. Then, open the tap and measure the response time until the disturbance has been eliminated, which in this case is considered to occur when the level stays within the interval from 39 to 41. Write down the response time in the table below. Repeat the experiment with integral time $T_I = 10, 5$ and 2 and write down the results in the table.

T_i :	Time:
20	
10	
5	
2	

What T_I corresponds to the shortest settling time?

Qualitatively, how does the response time depend on T_I ?

.....

Task 5.4

Close the tank tap. Choose $K = 1$, $T_I = 30$ and keep the reference value at 40. Repeat the experiment in the previous task, but now for the lower tank (make sure that **Tank2** is controlled and displayed). Write down also the minimum and maximum levels that lower tank reaches while the tap is open.

T_i :	Minimum/Maximum level:	Time:
30		
20		
15		
7		

What T_I corresponds to the shortest response time?

What T_I corresponds to the smallest deviation from 40?

Qualitatively, how does the response time depend on T_I ?

.....

If any of the assistants are available, then this is a good opportunity for partial reporting of the investigation. Otherwise, continue working and report later.

Partial report accepted by:

6 PD controller

For a PD-controller the relationship between the control error and control input is

$$u(t) = K \left[e(t) + T_D \frac{d}{dt} e(t) \right]$$

The corresponding parameters K and T_D can be chosen on the computer screen.

Differentiation of the reference signal is normally avoided. Instead the relation

$$u(t) = K \left[e(t) - T_D \frac{d}{dt} y(t) \right]$$

is used. Since $e(t) = r(t) - y(t)$ this makes no difference when the reference signal is constant. On the other hand, if the reference signal varies rapidly from one level to another, then the latter relation yields a less aggressive behavior in the control input.

Task 6.1

Recall from Task 4.2 that the lower tank was difficult to control with a pure proportional controller due to oscillations and long settling time. Here, we experiment with PD control to try to improve the response.

1. Remove the I-part by setting $T_I = \text{Inf} (\infty)$.
2. Click the button **I Reset**.
3. Select **P-part** and **D-part** in displayed signals.
 - Choose $K = 10$ and $T_D = 2$, and change the reference from 40 to 45 with tap closed, as in Task 4.2. Measure rise time $T_r = \dots\dots\dots$, overshoot $M = \dots\dots\dots$, and settling time $T_s = \dots\dots\dots$
 - Then, increase T_D to 10 and repeat the experiment and measure rise time $T_r = \dots\dots\dots$, overshoot $M = \dots\dots\dots$, and settling time $T_s = \dots\dots\dots$

What is the effect of differentiation on the step response? (compare with the pure P control in Task 4.2).

.....

.....

.....

- Explain how the D part works.

.....

.....

.....

.....

- What happens with the control signal for large T_D ? Why?

.....

.....

.....

7 PID control

A PID controller is obtained by introducing both integral and differential effects

$$u(t) = K[e(t) + \frac{1}{T_I} \int_0^t e(\tau) d\tau + T_D \frac{d}{dt} e(t)]$$

We now consider how different parameter choices in a PID controller affects the closed-loop behavior. Since it is time consuming to wait until all signals are constant, we will perform the experiments on a simulated two-tank system.

Task 7.1

1. Under **System settings**, turn the first switch to **Stop**, and then turn the second switch to **Simulation**.
2. Under **Controller settings**, choose **PID** and **Tank2**.

The simulation is now ready to start. The simulation speed can be changed in the **Simulation speed** box under **System settings**.

Study reference value changes from 40 to 45 for the lower tank. Start with $K = 6$, $T_I = 80$ and $T_D = 20$ and write down the results in the table below. Change the PID parameters and try to make the settling time T_s as small as possible with an overshoot M that is less than 5%. Write down the result when you find it satisfactory.

NOTE: Change only one parameter at a time; otherwise, you will not understand the effect of each parameter!

K	T_I	T_D	M	T_s
6	80	20		

8 Summary

Task 8.1

Answer the following questions with the recently achieved experience of PID controllers. Use the simulation tool if necessary.

- Increasing the gain K for a P controller typically has the effects

.....
.....
.....

- Decreasing T_I (increasing $1/T_I$) for a PI controller typically has the effects

.....
.....
.....

- Increasing T_D for a PD controller typically has the effects

.....
.....
.....

- Are the conclusions above valid for a PID controller as well? Motivate!

.....
.....
.....